

62. Graphics Subsystem

62.1 Graphics Overview

The graphic domain has the following features.

- Graphics LCD Controller (GLCDC): 8/16/18/24-bit RGB LCD interface supported.
- 2D Drawing Engine (DRW): Raster and vector graphics supported.
- MIPI D-PHY controller (MIPI PHY): Up to 2Lanes.
Max rate: 720 Mbps/Lane
- MIPI Display Serial Interface (MIPI DSI): Display Serial interface 2 supported.
- MIPI Serial Interface (MIPI CSI):
MIPI Alliance Specification for Camera Serial Interface 2 (CSI-2) Ver 3.0
MIPI Alliance Specification for D-PHY Ver 2.5 (Supported functions are equivalent to Ver 2.1).
- Video Input Module (VIN): Stores YCbCr-422 data and RGB data in memory through MIPI CSI interface.

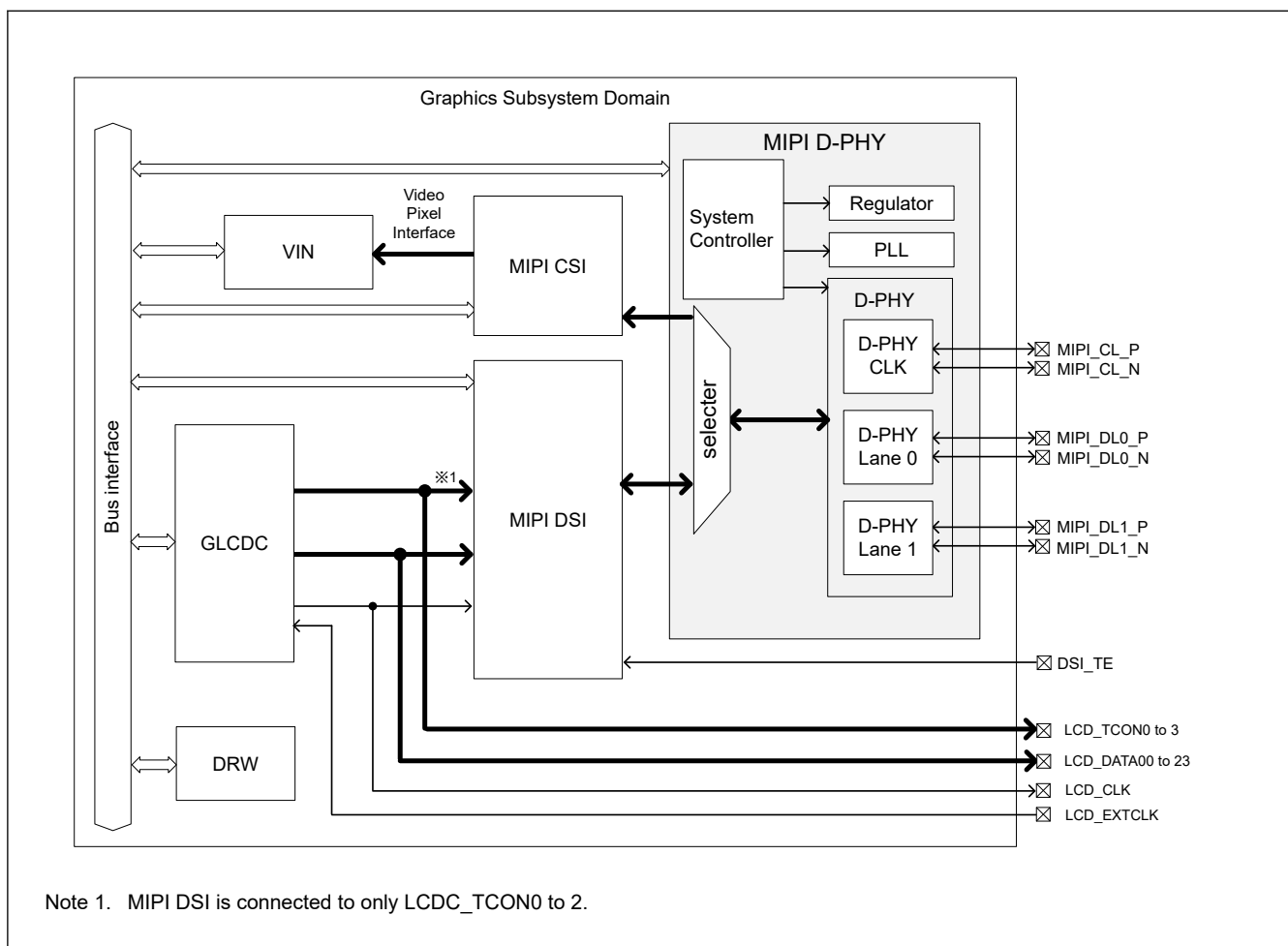


Figure 62.1 Graphics domain block diagram

Table 62.1 MIPI subsystem I/O pins (1 of 2)

Pin name	I/O	Function
MIPI_CL_P	DSI: Output CSI: Input	DSI/CSI Clock Lane positive pin
MIPI_CL_N	DSI: Output CSI: Input	DSI/CSI Clock Lane negative pin

Table 62.1 MIPI subsystem I/O pins (2 of 2)

Pin name	I/O	Function
MIPI_DL0_P	DSI: I/O CSI: Input	DSI/CSI Data Lane 0 positive pin
MIPI_DL0_N	DSI: I/O CSI: Input	DSI/CSI Data Lane 0 negative pin
MIPI_DL1_P	DSI: Output CSI: Input	DSI/CSI Data Lane 1 positive pin
MIPI_DL1_N	DSI: Output CSI: Input	DSI/CSI Data Lane 1 negative pin
DSI_TE	DSI: Input	DSI Tearing Effect pin
AVCC_MIPI	Power	D-PHY Analog Power Connect this pin to VSS_MIPI by a 0.1-μF capacitor. The capacitor should be placed close to the pin.
VCC18_MIPI	Power	D-PHY I/O Power Connect this pin to VSS_MIPI by a 0.1-μF capacitor. The capacitor should be placed close to the pin.
VSS_MIPI	Power	D-PHY GND

62.1.1 Start and Stop MIPI DSI and MIPI CSI

It is recommended that the settings for starting and stopping MIPI DSI and MIPI CSI be performed in the order shown in [Table 62.2](#) below.

Table 62.2 List of Operations

Function	Item	No.	Description
MIPI DSI	Start setting	1	See Startup in section 64.3.1. Overall Control .
		2	See section 65.3.1. D-PHY Startup Procedure .
		3	See section 66.3.3. Start of HS Clock .
	Stop setting	1	See section 66.3.4. Stop of HS Clock .
		2	See section 65.3.2. D-PHY Stop Procedure .
		3	See Stopping and restarting in normal operation or Stopping and restarting in abnormal operation in section 64.3.1. Overall Control .
MIPI CSI	Start setting	1	See section 65.3.1. D-PHY Startup Procedure .
		2	See section 68.3.11. Initialization Procedure .
		3	See section 67.4.1. Start/Stop of Reception Operation .
	Stop setting	1	See section 68.3.12. Capture Stop Procedure .
		2	See section 67.4.1. Start/Stop of Reception Operation .
		3	See section 65.3.2. D-PHY Stop Procedure .